PUPPYPI · DEEP RESEARCH · 31 OCTOBER 2026

PuppyPi room intelligence: temporal evidence, predictive models and active inspection

Temporal evidence, predictive observation and a measured path to an impressive room assistant. Proposal and literature review. Physical PuppyPi capability remains unverified.

27	6	84	0
SOURCES	FINDINGS	TRACKED CLAIMS	HARDWARE TRIALS IN THIS REVIEW

Research date: **2026-10-31**. Mode: **Deep**, using the eight-phase Deep Research skill. Platform: user-confirmed PuppyPi Standard, host computer, optional fixed phone. **Status: proposed research and application design; no new hardware capability.**

This report extends the twelve-work audit and revises its research priority. Read the current feature inventory for actual delivery status. The source registry, evidence store, claim ledger and run manifest make the review inspectable. Numbered citations identify works, not independent confirmations from different pages belonging to the same project.

Executive Summary

The strongest project direction is a room assistant that can explain both what it observed and what its observation history cannot establish. This review finds direct precedents for long-term temporal question answering, choosing between memory retrieval and physical exploration, uncertain memory refinement, perception diagnosis, and predictive inspection. Those precedents substantially narrow the earlier proposal's novelty claim. In particular, Mind Palace reports a false desk-use conclusion caused by sparse patrol coverage, while AbstainEQA already includes temporal information unavailability. **[1] [4]**

The recommended application is **PuppyPi Room Observatory**: taught stations, object identity, an evidence timeline, change investigation, bounded patrol and an optional external phone view. Its distinctive demonstration would let an evaluator choose an intervention and a time-qualified question. The app would show the observation gaps, select a useful measurement where possible, and return a supported answer or a precise unresolved result. For future monitoring, it would allocate robot and phone captures to reduce question-relevant gaps.

The proposed research question is whether task-conditioned acquisition improves supported-answer coverage at a fixed false-claim rate and sensing budget over periodic sampling with conservative temporal monitoring. This is a hypothesis, not certified algorithmic novelty. Partial-trace temporal

semantics, active diagnosis and joint reliability estimation are established; they are foundations and required baselines. **[8] [9] [12] [26]**

Learned world models and dense reconstruction remain optional host experiments. Released manipulation models do not supply PuppyPi motor control, and some strong RGB change methods have substantial offline computation costs. **[5] [19]** The first implementation should therefore establish evidence provenance, visibility-qualified temporal claims and replay, then add phone acquisition and measured action-outcome prediction. A counterfactual-history benchmark, explicit assumptions, independent ground truth and causal alarm timing can make the work academically useful even if simple baselines win. No upstream model was installed, paid API called, or robot actuated during this review.

Introduction: question, scope and methodology

The practical question is broader than how to attach a language model to a robot: what room understanding would make a small quadruped useful, visually compelling and scientifically assessable? This report considers a user who wants both demonstrable features and R&D, has more than three months, and already has a Python evidence core and an English-first mobile console. Those are project constraints, not externally measured research results. The application should reuse that work rather than create a second isolated controller or demo database.

Three independent retrieval branches examined temporal memory/world models, active diagnosis/asynchronous observation, and platform/inspection benchmarks. A fourth pass examined foundational world models, reconstruction and temporal monitoring. Primary method and evaluation sections were read for the closest works; repository trees, selected implementation files and deployment notes were checked where available. The registry records narrower reading scopes when only an abstract or official implementation document was inspected. No source was treated as reproduced merely because its repository exists.

The initial framing asked whether joint camera reliability and object-change inference might provide a contribution. Retrieval exposed older diagnostic-graph, robust-estimation and asynchronous-fusion precedents. A second discovery was more actionable: a reported failure of temporal room understanding caused by incomplete observation coverage. The outline was consequently refined toward temporal claim support and prospective acquisition, with source diagnosis retained as a nuisance state and baseline. That refinement is an interpretation of the evidence, not an assertion that prior methods cannot be extended. **[1] [8] [10] [12]**

The review includes academic papers, manufacturer specifications and official software documentation. Paper, project page, code and dataset belonging to one work form one scientific independence cluster. Broad conclusions are checked against multiple works; a paper-specific fact is explicitly single-source and does not become externally replicated because three links repeat it. Numeric source scores in the registry are the skill's uncalibrated triage heuristic. They are not probabilities of truth, and they do not replace assessment of experimental design, privileges or reproducibility.

Material assumptions remain open. Standard does not identify the user's exact board, firmware, camera or installed software. Its current manufacturer packing list must be distinguished from options shown across the product family. **[15]** Host GPU availability, repeatable robot pose, external-

camera timing and reliable obstacle sensing are unverified. Accordingly, the proposal begins with taught regions and supervised, measured station actions. Dense metric navigation, continuous high-rate observation and a generic learned locomotion controller are outside the initial claim.

The success criterion for this research phase is a decision-ready design with counterevidence, testable predictions, known reuse constraints and a clear falsifier. It is not a promise of publication. The report distinguishes observed source facts from our proposed system, logical examples and experimental plans. All quantities proposed for future measurement remain proposed, and all existing physical results belong to their original platforms and experimental settings.

Main Analysis

Finding 1: temporal room understanding already exists; historical coverage is the harder boundary

Mind Palace already combines episodic scene-graph memory, temporal questions, exploration and value-of-information retrieval stopping. Its evaluation includes simulated and real scenes. Most relevant here, its limitations describe a desk-use conclusion that was wrong because periodic visits missed activity between observations. This is a direct motivation for studying evidence coverage rather than advertising long-term memory itself as a new capability. **[1]**

STAR similarly joins temporal retrieval and spatial actions in one decision loop. Its source contains memory components, but the inspected benchmark README remained forthcoming and licensing was not detected. Its simulation execution also uses privileged object masks, including in the reported realistic memory setting. Reuse therefore needs a clearly named common-interface adaptation rather than a claim of equivalent raw-RGB robot performance. **[2]**

DynaMem supplies another useful memory reference: registered RGB-D supports stale-geometry revision and grounded query cross-checking. Its dynamic grounding benchmark isolates memory queries from navigation, exploration and manipulation. **[3]** These distinctions matter because an apparent improvement can result from better input geometry, privileged detections or a narrower task. The PuppyPi comparison must separately identify what information each method receives and what action is actually measured.

Abstention also has a close predecessor. AbstainEQA explicitly includes events outside the observed timeframe as information unavailability. Its analysis of prompting and text-only controls warns against apparent refusal performance that depends on question wording rather than available visual evidence. **[4]** Our inference is that a useful new test must hold wording constant while changing observation history. A model should not earn credit merely for learning that questions containing a particular word are often unanswerable.

UQ-DAAAM refines uncertain semantic memory using multiview caption consistency. Its guarantees depend on stated score distribution assumptions; they concern uncertainty reduction rather than calibrated answer correctness. Refinement selects recorded views, with the reported pipeline doing this at shutdown, rather than demonstrating physical next-view navigation. **[7]** Caption agreement and temporal completeness are therefore different evaluation targets. A perfectly consistent description can still be unsupported for an interval the camera never observed.

Consider a case observed on shelf A at 10:00 and 10:10. One possible history leaves it there throughout. Another moves it to B at 10:03 and returns it at 10:07. The two recorded shelf images can be identical. “Where was it last observed?” can have a supported answer. “Did it remain there for the whole ten minutes?” cannot be established from those images alone. This is our logical example, not a measured result from any paper. Adding an attractive room graph does not introduce the missing observation.

That example changes the product requirement. An evidence timeline needs unobserved intervals, source validity and the scope of a claim, not just a row of successful sightings. Natural-language answers should distinguish a historical witness, a present observation and a model-based forecast. A present look can verify current placement; it cannot reconstruct an unrecorded excursion. A stored phone recording or a human witness may provide new historical evidence, but each must be identified as a separate source with its own reliability.

The resulting opportunity is deliberately narrower than “robots should know when they do not know.” The question is whether physically measured visibility, timing and identity constraints can improve which temporal claims a system supports, and whether prospective robot/phone acquisition can improve that support efficiently. A deterministic coverage checker is the strongest initial competitor, not an inconvenience to be omitted. If it matches a richer method, the appropriate outcome is a simpler implementation and a useful negative result. Further algorithmic novelty would require an advance beyond these existing memory, abstention and uncertainty frameworks.

Finding 2: a useful world model needs a declared prediction target and a deployment budget

For this project, use three separate terms. A **historical room model** stores supported identities, places and observations over time. A **predictive observation model** estimates what a permitted sensing action might reveal. A **learned environment simulator** predicts future latent states or images from actions. These are proposed architectural categories, not a claim that the literature uses one universal taxonomy. They prevent a scene graph, a renderer and a learned dynamics model from being evaluated as if they were interchangeable.

The foundational World Models work demonstrates learned recurrent representations and controller training through imagined environment dynamics. **[23]** DayDreamer extends world-model learning to physical robots, including a quadruped; its released implementation has robot-specific actor/learner configurations. **[24]** These are important precedents, but their existence does not establish compatible PuppyPi proprioception, actuator interfaces or online-learning conditions. Replacing the current motion boundary would be a separate hardware/control project.

V-JEPA 2-AC makes the action-domain problem concrete. Its studied controls are end-effector commands for Franka manipulation, and its specified planning comparison reports 16 seconds per action on an RTX 4090. **[5]** That timing belongs to its reported configuration, not every use of its encoder. The implication for PuppyPi is to evaluate optional frozen features or a small observation predictor offboard, while preserving existing bounded motor skills. The released action head must not be interpreted as quadruped velocity or servo commands.

WorldMem addresses long-term consistency of generated scenes with pose/time-indexed memory, evaluated through generation metrics in Minecraft and posed real imagery. **[6]** This is a different objective from proving that an enrolled physical case remained at a station. An imagined view may

help visualization or supply a candidate prediction; it is not captured evidence. The app must preserve that distinction even when the predicted image looks convincing. A synthetic room replay likewise cannot be presented as a recording of an actual patrol.

Reconstruction offers a useful visual extension. VGGT predicts cameras, depth and point/track geometry, with examples for COLMAP export and optional bundle adjustment. Its release distinguishes original and commercial checkpoint terms. **[25]** The proposed use is a host-generated room visualization registered against measured anchors. Before attaching metric claims to it, measure scale, pose and reconstruction error on held-out landmarks. A reconstructed surface is not by itself a certified traversable region, an object identity or a visibility proof.

SceneDiff is especially relevant as a training-free RGB spatial-change baseline. Its inspected implementation uses geometry, segmentation and pretrained features; the paper reports an average 159.4 seconds per sequence pair on an NVIDIA A40. **[19]** This supports an offline deep-audit role, not a promise of live per-frame processing on the Pi. Its paper, website and current evaluation revision also differ in reported headline gains and matching protocol. Freeze the revision and metric definition before making comparisons; a downloaded repository name is not a reproducibility specification.

A direct predictive-inspection precedent also exists: World Model Failure Classification and Anomaly Detection. It studies a fixed robot pose with PTZ control and offboard inference, using 290 short inspection videos. **[17]** That is strong motivation for a separate temporal-monitor comparison. It does not establish autonomous room traversal, raw sensing robustness or event coverage between sampled patrol stops. A successful gauge-monitor demonstration should not be transferred wholesale to free-moving object inspection.

Our proposed architecture consequently uses two rates. Cheap, bounded perception and source checks support live operation; selected episodes enter a queued host audit whose completion time is visible. An audit result refers to the evidence it processed, even if the robot has since moved. It can request a fresh check before a current-state report. These are design choices to be measured, not performance claims. The cost model includes upload, queueing, model inference and re-acquisition, rather than timing only one network forward pass.

For the first predictive experiment, forecast the outcome of a small set of verified observation skills: whether a region will be exposed, an instance feature discriminative, a tag usable and a capture fresh. Record the prediction before execution and score the measured result afterward. Compare a lookup table, a geometric/rule baseline and a small learned head using the same frozen visual features. Do not begin by training a large video generator. If predictions fail to improve task decisions after their cost is counted, the room assistant should remain a structured evidence system with transparent sensing policies.

Finding 3: joint reliability estimation and active diagnosis are foundations, not the remaining novelty

The previous proposal emphasized jointly considering a changed room and faulty camera evidence. That is a sensible state-estimation design, but the general pattern has older precedents. Switchable Constraints jointly optimizes geometric state and factor inclusion using latent switches. **[12]** PerceptionMonitoring models diagnostic relations, failure states and diagnosability; released source includes a Python/C++ toolchain with incomplete usage documentation. **[8]** Robust weighting and diagnostic graphs therefore belong in the baseline ladder.

Han et al. go further by predicting control-conditioned diagnostic outcomes and selecting controls through MCTS for fault isolation. Their evaluation is simulated monocular asteroid navigation. **[9]** Gao et al. compensate delayed object locations before collaborative fusion, assuming correspondences, extrinsics and delay estimates in an RGB-D setting. **[10]** These establish active diagnosis and asynchronous object estimation without proving that either transfers unchanged to an RGB quadruped and phone. An adaptation must preserve that distinction.

Timing is also an inference problem with observability conditions. Qin and Shen jointly estimate temporal offset with visual-inertial state and explicitly use sufficiently excited motion in their calibration experiments. **[11]** A static room and static external camera do not automatically offer such excitation. Likewise, single-frame deep view synchronization studies content alignment for unsynchronized multiview tasks with geometric assumptions; it is not sensor-freshness attestation. **[14]** Clock, image content and packet sequence are related evidence, not interchangeable guarantees.

Even a predict-test-update inspection loop is established. Sehr et al. use process uncertainty to guide robot sensing, while their hardware-in-loop experiment substitutes simulated process measurements for captured thermal imagery. **[13]** This is a useful warning about evidence labels: physical motion plus simulated perception is a mixed experiment, not a complete physical vision validation. PuppyPi reports should separate replay, synthetic observations, selected-view acquisition and actual closed-loop execution with equally explicit boundaries.

The sharper design target is **answer identifiability**. Suppose the phone may be stale, but every plausible current-world hypothesis agrees that a newly observed case at B is the taught instance. A task asking for current location may already be answerable without proving the phone's exact failure mode. Conversely, a task asking whether the case ever left A remains unresolved even if camera health is now completely known. Resolving all fault entropy can waste effort that has no effect on the requested answer. This is our interpretation and proposed policy objective, not an established new result.

A useful state representation therefore preserves combinations. An object can move while the phone freezes; a camera can move while two look-alikes exchange positions. A single categorical label such as change-or-fault excludes valid worlds. Source hypotheses should influence which observations are admissible and how uncertain they are, while world hypotheses describe placement and identity. The system need not diagnose an unseen fault with certainty before reporting a well-supported fact from an independent valid source.

There is an important impossibility boundary. If a static live camera and a frozen image produce exactly the same accessible observations, pixel equality does not identify which is present. A new packet timestamp only identifies receipt unless its relation to capture is established. A validated visual challenge, preserved historical recording or independently observed landmark motion can add information. If no feasible test separates answer-changing hypotheses, further repeated inference cannot turn disagreement into knowledge. The correct outcome remains unresolved, possibly with a grounded help request.

The experiment should first test deterministic age/session/calibration gates, visibility-qualified negatives and robust residual weighting. Next compare passive and active diagnosis cascades against task-conditioned acquisition using the same observations and permitted actions. A larger joint estimator is justified only if its end-task benefit survives those controls. MCTS is optional after a greedy selector; algorithmic complexity is a cost and an experimental variable, not evidence of research depth by itself.

Finding 4: temporal claims need explicit semantics before an intelligent acquisition policy can be evaluated

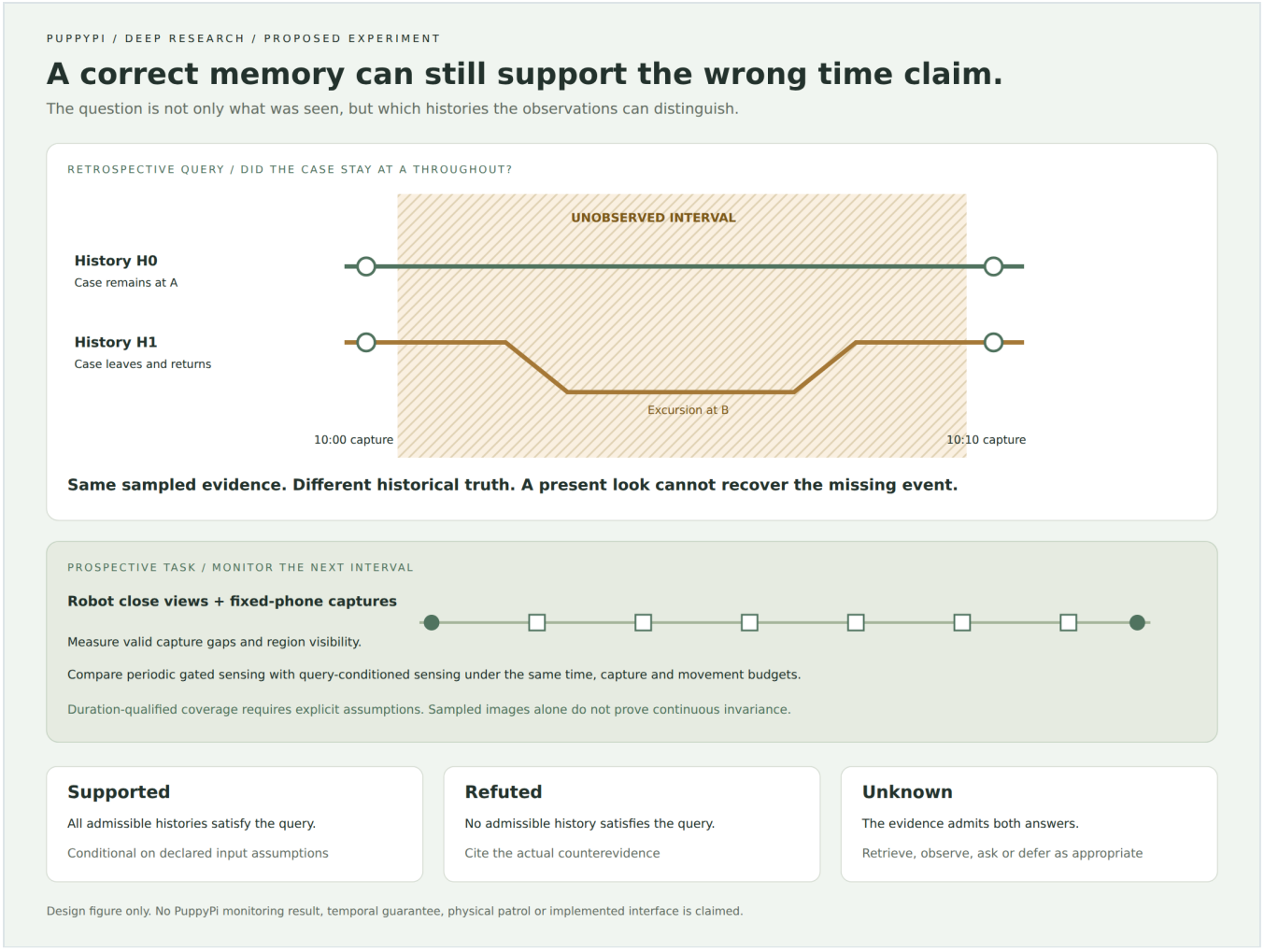


Figure 1. Proposed counterfactual-history test, not collected data or a delivered app screen. The phone row describes a future monitoring experiment, not recovery of missing past observations. Editable SVG.

The proposed method starts with a bounded query contract. Parse a request into an enrolled identity, a named region, a time horizon, a predicate and a temporal operator. Preserve distinctions such as last observed, present now, observed at every check, continuously present, or an excursion lasting at least a declared duration. Unsupported requests require clarification. The language model may propose this typed query, but it does not decide whether the evidence satisfies it and does not issue movement commands. This is a design proposal.

Temporal monitoring itself is established. Robust Online Monitoring of STL defines satisfaction intervals for partial signals and states an interpolation assumption for reconstructing sampled signals. **[26]** RTAMT provides online/offline monitoring interfaces, including a bounded-future online fragment and optional C++ backend. **[27]** The research must not rename three-way monitoring as a new logic. The room specific problem is constructing defensible input signals from uncertain identity, geometry, source validity and capture time, then measuring whether selecting observations improves useful decisions.

Our proposed semantics uses a set of histories compatible with the accepted evidence and declared assumptions. Let an evidence record retain capture-time bounds, receipt time, source/session, model/preprocessor version, region, identity candidates, calibration reference and observed visibility. Let `H(E, A)` denote histories consistent with evidence `E` under assumptions `A`. For a typed query `q`, return Supported if every admissible history satisfies it, Refuted if none does, and Unknown otherwise. If `H` is empty, report inconsistent evidence and investigate admission/assumptions; do not accept every query through vacuous truth. This is a specification for a small prototype, not implemented code.

This set-valued view is deliberately separate from a probability distribution. It can establish an assumption-dependent support relation without inventing probabilities. A probabilistic forecast additionally needs a transition and measurement model fitted and calibrated on appropriate held-out episodes. “Likely remained at A under this learned model” is a forecast, not equivalent to “observed continuously at A.” The app should make that distinction understandable without requiring the operator to inspect equations.

Elementary indistinguishability argument. Take any finite set of valid robot capture times with a nonempty gap. Construct history H0 with the object at A throughout and history H1 with an excursion-and-return entirely inside that gap. Allow unrestricted object transitions and make all recorded views otherwise identical. Any method receiving only those observations gets the same input in both histories, although their continuous-presence and ever-left answers differ. It cannot always issue a correct definite answer for both. This is a simple logical limitation, not a novel theorem or an empirical failure rate.

The argument also explains why a retrospective query is not always an acquisition task. A fresh look may resolve present location or identify which stored crop belongs to the case. It cannot reveal a vanished past excursion without an additional witness or recording. The action selector must distinguish recoverable stored evidence, current-state ambiguity, irrecoverable past ambiguity and a future monitoring request. Otherwise it can spend a large movement budget on a question no future view can answer. Human help counts as a source with response time, provenance and possible error; it is not a hidden ground-truth oracle.

A narrower duration contract permits a useful conditional statement. Suppose every excursion lasts at least `delta`, source validity and object labels are correct at captures, the relevant region is observable, and valid captures span the requested horizon including its endpoints. If every actual adjacent gap is strictly shorter than `delta`, an excursion within that horizon cannot occur and return entirely unwitnessed: it would have to fit inside a shorter gap. Boundary crossings are witnessed at the endpoints. This elementary sampling argument is an explicit operating assumption, not a general guarantee that a camera detects all real events.

With uncertain capture times, use the largest admissible adjacent gap and a certified observed span. A nominal timestamp near the query boundary does not establish that the boundary itself was observed. Occlusion, uncertain identity, source failure, exposure blur or wrong labels invalidate the proposition's premises. Camera frame rate alone cannot certify continuous coverage, and intervals between accepted frames cannot silently become observations through constant interpolation. The initial benchmark should sweep event duration against capture gaps and report where this contract actually holds.

An implementable first version can maintain disjoint support, contradiction and unknown intervals for each enrolled object/region predicate, together with the assumption used to bridge any interval. It should preserve negative evidence only where the object could genuinely have been seen. A

detector miss from a low or blocked view does not prove absence on a tabletop. If geometry is insufficient, operator-specified region visibility may support a labeled experimental mode, but that mode must remain distinct from automatic inference. Exact-instance identity is evaluated separately from class recognition.

For prospective monitoring, define a finite set of feasible captures, station inspections and help requests. Our proposed objective minimizes future task loss plus measured sensing, travel, compute and operator costs. A deterministic policy may minimize the largest query-relevant blind gap; a stochastic policy may use an explicitly estimated event model to minimize expected loss. Neither objective follows from a set of admissible histories alone. Include a no-action/defer option, remaining budget and action failures rather than assuming each command creates a useful observation.

Adaptivity needs a reason to help. In a homogeneous single-region setting with equal costs and no useful event prior, evenly spaced valid observations are a strong baseline for controlling the maximum gap. The study should include that setting and expect little advantage. More informative conditions have multiple query regions, complementary ego/exo visibility, changing occlusion, variable transmission/inspection cost or measured event context. The policy must exploit observable differences rather than private intervention schedules. If benefit exists only when supplied true hidden visibility or event timing, it does not support the deployable research claim.

The more ambitious model predicts the probability that a permitted action produces an admissible, answer-changing observation. Its targets are identity discrimination, region visibility, timing validity and execution success, not beautiful imagined pixels. Train it on measured action outcomes, calibrate it separately and compare against empirical frequency tables. A useful ablation keeps all state and perception fixed while replacing query-conditioned utility with generic entropy or coverage utility. That tests the proposed decision mechanism instead of confusing a new detector, extra camera and new planner.

Finding 5: the benchmark must expose impossible answers and causal evaluation errors

The proposed benchmark has two complementary goals: evaluate supported temporal answers and evaluate how acquisition improves future observability. It must not reward a method for confidently guessing a hidden event from a biased question. The central unit is a mission with a query, admissible evidence, permitted actions, an intervention history and a separately protected reference record. Frames from one mission are correlated observations, not independent trials. All counts and splits below describe a proposed protocol, not collected data.

Start with paired histories. Use identical enrolled objects, room layout and question wording, and make the robot's sampled observations match as closely as possible. In one history an object stays at A; in the other it leaves and returns between visits. A retrospective arm supplies no additional historical witness. A prospective arm begins the monitoring request before the intervention and permits budgeted phone captures or station checks. The pairing separates evidence sufficiency from language shortcuts and tests whether the policy buys new information instead of inventing it. The strict indistinguishability replay arm must use byte-identical model-visible frames and relevant metadata for the paired histories. Approximately matched physical trials test robustness rather than the literal impossibility argument.

Cross that design with no-change, persistent move, excursion-return and look-alike exchange conditions. Vary view obstruction, capture gaps, phone delay, camera relocation and known/unknown timing. Include true changes together with source faults and episodes in which neither camera can resolve identity. Keep source-fault labels out of the method's inputs unless a baseline is explicitly given oracle information for diagnosis. Easy faults with trustworthy timestamps belong in the benchmark, but cannot be the only cases: a timestamp threshold would otherwise solve the advertised challenge.

There are three evaluation modes. Shared-evidence replay fixes all observations and perception outputs, isolating memory, admission and query semantics. Controlled acquisition lets each policy request real supported actions against repeated staged conditions, measuring the observations it actually obtains. Supervised physical missions add localization, movement, network and intervention failures. A fixed recorded trajectory cannot fairly evaluate a request for an unrecorded view. Synthetic branching can test policy logic, but its result must remain separately labeled rather than mixed into a physical success rate.

The strongest baseline is a partial-trace monitor with periodic acquisition and conservative source/visibility gates. Compare it with the same monitor using generic coverage/entropy acquisition, then query-conditioned acquisition using the identical state estimator. Add a current-state graph, a retrieval/VLM QA adaptation, and prompted abstention as application comparisons. Diagnosis-first cascades and robust weighting test whether the source model adds anything. Optional learned observation forecasting must face a measured lookup-table baseline. An oracle schedule or hidden visibility labels are diagnostic upper references only, never deployable competitors.

The primary result should be supported-answer precision against decision coverage at matched mission time and sensing budgets. Include event recall, unresolved true changes, false definite answers on indistinguishable pairs, capture-gap distributions and latency from event onset to the **first causal report**. Universal-claim errors and sampled-check answers require separate scoring. A system returning Unknown everywhere avoids false assertions but fails coverage and recall; a system answering everything can have misleading average accuracy if easy unchanged episodes dominate. Report the complete tradeoff.

Maintain two separate evaluation labels: hidden scene truth and evidence-relative support under the declared query contract. A lucky claim that the case stayed at A in the unchanged history is still unsupported if the accessible evidence also admits the excursion history. Both members of that retrospective pair warrant Unknown for the universal question. Score factual correctness and evidential support separately. For the bounded task set, adjudicate support from accessible evidence, documented assumptions and counterhistories; record reviewer disagreement. Do not label a claim supported merely because the protected reference recording happens to make it true.

There is a concrete reproduction hazard in the inspected inspection timing script: the default chooses a detected spike nearest the ground-truth label, while first detection is an option. This is not proof of the setting used for the paper's numbers. **[17]** For PuppyPi, predeclare causal first-report timing, early false alarms, event-matching windows and repeated-alarm handling. Ground truth may score outputs afterward; it may not select the most favorable deployed alarm.

Physical reference design also needs care. LILocBench reserves an upward camera for AprilTag ground truth rather than estimator input. **[20]** AutoInspect reports missions, completed inspection actions and interventions over an extended physical deployment. **[16]** Adopt those accounting principles without transferring their sensors, platforms or success rates. For this project, record

actual robot pose independently where possible and count every failed acquisition, operator correction and incomplete return. Reference markers cannot secretly leak identity or pose labels to a method supposedly evaluated without them.

Group train/calibration/test by room, day, object instance or intervention family as the eventual claim requires. Preserve reference/rescan associations and reject near-duplicate frame leakage. Fit event likelihoods and action costs only on the training split; choose thresholds on calibration; freeze them before held-out evaluation. A pilot estimates variability and informs sample-size planning. Confidence intervals should resample missions or room/day clusters, not adjacent video frames. If only one room is available, state same-room robustness rather than unseen-room generalization.

Finally, test whether the claimed acquisition effect survives cost accounting. Report equal wall-clock, equal accepted-capture and equal inference-budget comparisons, with actual bytes, queued-audit time, travel and operator response time. Larger sensing opportunity is an application advantage but not evidence that an algorithm uses the same opportunity better. A convincing negative result would show that periodic gated captures match a complex policy under realistic conditions; that result should change the final implementation rather than be hidden behind a weak fixed-patrol comparison.

Finding 6: the impressive application is an observable investigation, supported by modest hardware first

The application proposal is **PuppyPi Room Observatory**. Its visible center is a room map linked to time: places, enrolled objects, observed changes, camera coverage and unresolved questions. The name describes an intended product flow, not a delivered feature. It extends the existing field console and shared evidence store. The current application still uses operator assertions and a synthetic default robot; autonomous room perception, phone ingestion and physical patrol require their own integrations and acceptance evidence.

The manufacturer specification supports a conservative starting point. The Standard packing list is narrower than the product family's combined options: ranging/voice equipment appears in higher kits. Board configuration must be physically inventoried. **[15]** Begin with RGB, host computation, taught regions and measured routes. A phone can add an external observation, but cannot make a low camera see every table or provide general collision avoidance. The mission must stop or defer when its validated observation or movement conditions fail.

External localization has prior art with substantial assumptions. Bultmann et al. use calibrated external cameras with robot-specific pose estimation and an already sensor-equipped mobile platform. **[18]** For the initial PuppyPi experiment, use rigid fiducials and independent reference poses before a learned external pose estimator. AprilTag requires camera calibration and correctly measured tag geometry. **[21]** Tags can support pose checks; they are not object-instance recognition or obstacle sensing.

Phone transport can build on the aiortc browser/server example, which provides reusable webcam media exchange. **[22]** The project still needs source/session identity, capture-time bounds, bounded buffering, calibration lifecycle and reconnect handling. Browser permission and transport arrival do not establish pixel freshness. A phone used as the operator display is also not automatically a simultaneous observer; device placement, power, capture support and actual browser behavior need a separately measured setup.

On the host, keep the pure Python evidence, temporal monitor and task policy independent of ROS, UI and model transport. Optional perception/reconstruction adapters convert their outputs into project-owned records, with explicit model and preprocessor identity. Heavy GPU stacks should use separate optional environments or services with versioned input/output contracts. A failed deep audit must not stall the watchdog or prolong motion. The Pi retains the existing vendor gait/control boundary; physical gait and stop behavior require independent validation on the actual kit. Neither a VLM answer nor a generated video supplies motor instructions.

The first new durable records would describe a query, admitted observations, source/calibration revisions, predicate support and proposed/measured action outcomes. Their names are design choices, and persistence requires an explicit schema migration. Preserve the original image evidence and revisions rather than overwriting an old location when a new belief arrives. Delayed evidence can revise a historical claim through a bounded recomputation with an audit trail; it cannot silently move the receipt timestamp into the capture slot.

The demo begins with enrollment and a few named room regions. The evaluator selects a case and asks a retrospective question about the last interval. A timeline shows accepted robot views and gaps. If the case briefly moved and returned in an unobserved gap, the correct explanation identifies that missing evidence instead of inventing a movement story. The evaluator can reveal the reference recording afterward. This demonstration is valuable only if the model never received that recording or its labels before answering.

Next the evaluator starts a prospective monitoring task. The app previews which regions are currently observable and how a phone capture or robot station visit could improve the evidence. The policy selects an allowed action, logs its expected outcome and cost, and displays the actual result after acquisition. An enrolled case and its similar decoy require a close discriminative view. Moving the phone invalidates its geometry until checked; a source fault can coexist with a real move. These are proposed behaviors to implement and test, not a script of predetermined final answers.

The user experience should expose meaningful outcomes without presenting the operator with internal model machinery. A Room view shows locations and evidence age. A Timeline distinguishes observations, unobserved spans and forecasted coverage. Investigate shows a concise reason for the next permitted measurement. An answer links to its supporting frames and states whether it concerns current placement, observed checks or a duration-qualified monitoring task. Replay shows what was predicted, executed and measured; export includes source provenance. These views provide feature breadth while sharing one evidence model.

If host reconstruction succeeds, add an optional registered 3D view alongside the measured station map. If deep spatial audits are useful but slow, show them as queued reports with their original capture interval. If no feasible action can resolve a query, offer a grounded help request or retain Unknown. Pause, manual takeover, explicit revalidated resume and the measured HOME outcome remain separate execution requirements. A visually compelling demo should make those boundaries visible through actual behavior, rather than hiding unresolved perception or motion behind a polished animation.

Synthesis & Insights: a sharper research proposal

The recommended provisional title is **Coverage-Aware Temporal Room Inspection with Active Ego–Exo Observation**. It names an investigation rather than declaring a new foundation model. The broad architecture combines established memory, monitoring and active-perception ideas. The possible contribution is a measured method or benchmark that ties temporal claims to the observations capable of supporting them, with prospective acquisition under real visibility and cost. Publication novelty remains unresolved and would require a focused review of the final method, not just this application direction.

Three hypotheses separate the possible benefits. H1 concerns integrity: visibility/time/source-qualified temporal semantics reduces false definite answers on incomplete histories compared with latest-state or prompted QA systems. H2 concerns acquisition: query-conditioned robot/phone sensing yields more supported answers than periodic gated sensing under the same budget when visibility and costs are heterogeneous. H3 concerns prediction: a small learned action-outcome model improves that tradeoff beyond empirical tables. Each can fail independently. H1 may succeed while H2 shows periodic observation is enough, and H3 may add cost without useful improvement.

Do not call all three contributions delivered by one successful demonstration. The first experiment should isolate H1 in shared replay. The second begins the query prospectively and isolates H2 with identical perception. The third keeps the acquisition objective fixed and substitutes learned predictions for tables. Optional dense reconstruction and expensive spatial audits are application or representation studies with separate measurements. This structure makes it possible to retain useful features while rejecting a research hypothesis.

The question's temporal operator changes what actions have value. A current location query can benefit from a new close view. A past event query benefits from searching preserved recordings or witnesses. A continuous-interval query may require assumptions the system cannot justify. A future duration-qualified monitoring task can benefit from controlling observation gaps. Therefore, a policy that always selects the view with greatest generic uncertainty reduction can optimize the wrong problem. This is the proposed mechanism to test, not a measured advantage over the cited systems.

The phone's role is similarly conditional. It may provide long observation duration at a fixed pose, while the robot supplies close identity evidence or access to another region. It may also be occluded, stale, geometrically invalid or aimed at the wrong region. Treat its contribution as a measured change in query-relevant observability, with bandwidth and processing costs. “Two cameras” is an equipment count; “fewer unsupported temporal answers at equal cost” is a research outcome. If a fixed phone alone solves the monitoring task, report that result and avoid attributing the gain to quadruped planning.

The highest-value reusable output could be the benchmark even if no elaborate policy wins. A useful release would contain selected cleared clips, immutable run manifests, identical-wording query pairs, protected reference labels, explicit timing/visibility uncertainty and baseline adapters. It would include unanswerable episodes and intervention counts instead of filtering them away. That output can support future work on memory admission, temporal inference, active sensing and observation forecasting without requiring a particular model provider or robot controller.

The minimum claimed advance must be stated after the results. A system paper could contribute a carefully characterized low-cost deployment and open protocol. A benchmark paper would need evidence that its tasks reveal reproducible gaps not explained by lexical cues or privileged inputs. An algorithm paper would need a precise mechanism and strong matched ablations. Hardware affordability, framework count, a scene graph plus chatbot, or the use of a recent pretrained model are not sufficient arguments for algorithmic novelty. This report is intended to prevent those claims from entering the final project narrative.

Counterevidence Register

Attractive claim	Counterevidence or strongest alternative	Required response
Temporal memory and active room QA are new	Mind Palace and STAR already couple memory and action **[1] [2]**	Reuse/adapt and compare; narrow the question
Abstention for missing past evidence is new	AbstainEQA includes temporal unavailability **[4]**	Evaluate observation-dependent behavior with fixed wording
Joint source reliability is new	Diagnostic graphs, robust switches and active FDI **[8] [9] [12]**	Include gates, robust weighting and diagnostic cascades
A predictive video proves room understanding	Generation, manipulation and inspection have different tasks **[5] [6] [17]**	Score physical task decisions and preserve imagined/captured provenance
Temporal intervals guarantee continuous truth	Partial-signal monitoring requires explicit signal assumptions **[26]**	State admissible histories and measure input validity
Adaptive sensing must beat periodic capture	Homogeneous costs/visibility may offer no adaptive advantage	Include a null setting and retain the simpler policy if matched
The camera setup is lightweight because the robot is small	Reconstruction/audit models have independent host costs **[19] [25]**	Measure end-to-end time and label queued work

Claims-Evidence Table

The machine-readable ledger links atomic facts and synthesis statements to stable source/evidence IDs. This table records the main conclusions and their status; it does not count project pages as independent experimental replication.

Conclusion	Independent work clusters	Status
Generic memory, temporal QA and semantic refinement have close predecessors	1, 2, 3, 4, 7	Supported scope assessment; no claim that their methods are identical
Generic joint reliability, async fusion and active diagnosis are established	8, 9, 10, 11, 12, 14	Supported; sensor/task assumptions differ
Learned dynamics, visual simulation and inspection models are not interchangeable deployment claims	5, 6, 17, 23, 24	Supported comparison plus project-specific inference
The proposed temporal/acquisition mechanism is novel	None establishes this negative universal	**Unverified; not asserted**
Query-conditioned sensing improves the proposed benchmark	No PuppyPi experiment yet	**Hypothesis H2**
The actual Standard kit supports autonomous patrol	No physical acceptance evidence yet	**Unverified; prerequisite**

Limitations & Caveats

This is a targeted deep review, not an exhaustive systematic review with a claim of literature completeness. Searches were refined after counterevidence emerged; the final evidence set therefore emphasizes close mechanisms and deployability rather than a balanced census of every semantic mapping paper. Some sources are preprints. Reading depth and source-release checks vary and are recorded in the registry. A paper-specific finding is not an independent replication, and a repository snapshot does not establish successful installation or performance.

The benchmark and policies are unimplemented. No physical timing, visibility, motion or prediction accuracy is reported for PuppyPi. The proposed admissible history set is only as defensible as its inputs and assumptions. Incorrect perception or aggressive pruning can discard the true history and invalidate apparent support. A conservative model can produce too many unknowns. The study must expose both failure modes through precision/coverage and reference audits, rather than presenting conservatism as free correctness.

There may be closer work on active temporal monitoring, camera-network scheduling or evidence-aware EQA than the retrieved set. The final algorithm would need a follow-up review against its exact formulation. No first-of-its-kind claim is made here. Existing software licenses, checkpoint access and dataset conditions can change; revisions and terms must be checked again at integration time. The manufacturer's current configuration is an assumption until reconciled with the actual kit. Host compute and real phone behavior remain unresolved.

Automated report checks can verify structure, links and lexical evidence overlap. They cannot prove a source entails a scientific claim or that a causal inference is valid. The review therefore includes manual primary-section inspection and independent critique. Any network verification warning must remain visible in the validation record, even when the same source was accessible through the web retrieval tool. A successful validator exit is not a substitute for that audit.

Recommendations

Implement one coherent evidence pipeline before expanding the feature inventory. The first slice is a versioned run ledger, source lifecycle and replay that keeps capture bounds separate from receipt time. Add typed temporal queries and a conservative interval/check monitor with explicit unknowns. Demonstrate the two indistinguishable historical cases on replay. This is a reviewable software milestone that requires neither GPU models nor physical locomotion and provides a meaningful failure scenario from the start.

In parallel with software, inventory the board, vendor environment, camera, movement primitives and stop behavior. Establish stationary live capture before supervised station routes. Add fixed-phone capture only after measuring timing, calibration validity and source loss. Automatic visibility and exact-instance matching require labeled real observations; operator annotations should remain an explicit mode, not silently stand in for perception performance. Do not make reconstruction or a learned controller a prerequisite for the core app.

Use the first development episodes to compare periodic gated sensing and a query-conditioned greedy policy. Add heterogeneous regions and measured costs, then lock the benchmark protocol and held-out splits. An optional small learned observation predictor is justified only after empirical tables are competitive. Queued SceneDiff-style audits and a registered 3D viewer can add application value in a separate host experiment, with latency and provenance visible.

For a conditional sixteen-week schedule, allocate the first four weeks to hardware characterization, the ledger, perception labels and temporal replay; the next four to phone capture and strong fixed/active baselines; the next four to measured acquisition and the evaluator-driven app flow; and the final four to held-out trials, critique, figures and reporting. This is a capacity estimate, not a promise. If route validation fails, keep the study stationary or operator assisted. If a simple monitor matches the complex policy, ship the simple monitor and report why adaptivity did not help.

The final presentation should distinguish delivered features, integration evidence, hardware verification and research hypotheses. Its strongest story is a question whose answer changes when the available evidence changes, with the robot demonstrably choosing a useful measurement. A feature checklist can show application breadth, but the scientific conclusion must come from matched comparisons and honest failures, including unresolved cases.

Methodology Appendix

The invoked skill was retrieved at revision `f2f2c0fa4e7617ca84c86b63f4bb40f77a746933`, after checking the local skill catalog and discovering no installed Deep Research package. Its methodology, report-assembly, HTML and quality references were read. It was used as a retrieved workflow with reviewed validation scripts, not as a claim to have invoked a hosted Deep Research service. Source discovery used the available web tool because the skill's preferred search CLI was absent. Three parallel agents returned structured evidence; the root agent integrated, critiqued and refined it.

Scope and plan covered eight angles: temporal memory, long-term EQA, abstention, predictive world models, active diagnosis, asynchronous camera evidence, physical inspection/platform constraints and evaluation methods. Retrieval preserved short quotations and locators plus

bounded paraphrases in an append-only evidence file. Canonical work identifiers merge arXiv HTML/abstract/version URLs into one work cluster while retaining the inspected version in its URL. Related code and project pages do not become extra independent scientific support.

Triangulation tested the broad memory, diagnosis and world-model claims across multiple works. Specific reported numbers remain single-source and are linked to their experimental setting. The critique used practitioner, skeptical reviewer and implementation-engineer perspectives. It rejected generic novelty claims, identified the periodic-monitor baseline, and separated retrospective inference from prospective acquisition. The counterfactual-history example and elementary sampling argument are our logical analysis; their utility and assumptions still require empirical validation.

Each major section was written progressively from the persisted evidence. Atomic factual claims and synthesis/recommendation records are separately typed. The skill's report, citation, claim-support and HTML validators are run alongside repository document checks. Their results and manual-review qualifications are recorded in the validation artifact. Numeric heuristic overlap/credibility scores are not treated as calibrated probabilities. No whole paper, complete camera stream, downloaded weights or credentials are committed with this report.

The repository is the canonical documentation destination, following its existing format and English language rules. A readable HTML/PDF bundle is also prepared under the skill's Documents destination. Packaging includes a proposed design figure, not a screenshot of unimplemented features. Existing synthetic app screenshots remain separately labeled in the current gallery. The manifest lists artifacts and adaptations, and the handoff records the next concrete code slice without changing feature delivery status.

The stopping rule is evidence adequacy and completed critique for this scoped decision. More search results would not establish novelty by accumulation. Remaining uncertainty is actionable: reproduce the closest feasible baselines, measure the actual observation process, then decide whether any proposed coupling earns additional implementation complexity. That process can support a strong application and a credible R&D report even if the most elaborate method is rejected.

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Canonical Markdown and evidence ledgers live in the PuppyPi repository. Generated HTML/PDF are reading copies. Proposed behaviors are not delivered robot features.